

CLOUD COMPUTING

UE23CS351B

LAB 2- Introduction to Docker

GITHUB:

<https://github.com/akshayag2005/CLOUD-COMPUTING>

DATE: 27/01/2026

NAME: AKSHAY

SRN: PES2UG23CS044

SECTION: A

1. SCREENSHOT-1

```
(akshay@AKSHAY)~$ mkdir PES2UG23CS044
(akshay@AKSHAY)~$ export SRN=PES2UG23CS044
(akshay@AKSHAY)~$ echo "SRN: $SRN"
SRN: PES2UG23CS044
(akshay@AKSHAY)~$ docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/

(akshay@AKSHAY)~$ |
```

2. SCREENSHOT-2

```
(akshay@AKSHAY)~$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
(akshay@AKSHAY)~$ docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
014c3a9ef275   hello-world   "/hello"   6 minutes ago   Exited (0) 6 minutes ago   great_darwin
2494e30e04eb   hello-world   "/hello"   11 minutes ago   Exited (0) 11 minutes ago   wonderful_sammet
(akshay@AKSHAY)~$ docker images
REPOSITORY    TAG       IMAGE ID   CREATED   SIZE
python        3.10-slim f5d029fe3914   2 weeks ago   185MB
hello-world    latest    05813aedc15f   5 months ago   25.9kB
(akshay@AKSHAY)~$ |
```

3. SCREENSHOT-3

```
(akshay@AKSHAY)~$ docker pull python:3.10-slim
3.10-slim: Pulling from library/python
Digest: sha256:f5d029fe39146b08200bcc73595795ac19b85997ad0e5001a02c7c32e8769efa
Status: Image is up to date for python:3.10-slim
docker.io/library/python:3.10-slim

(akshay@AKSHAY)~$ docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
python               3.10-slim          f5d029fe3914       2 weeks ago        185MB
hello-world          latest             05813aedc15f       5 months ago       25.9kB

(akshay@AKSHAY)~$ |
```

4. SCREENSHOT-5

```
(akshay@ AKSHAY) - [~/PES2UG23CS044/ml-docker-PES2UG23CS044]
$ docker build -t spam-web-app-pes2ug23cs044 .
[+] Building 43.0s (12/12) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 752B                                0.0s
=> [internal] load metadata for docker.io/library/python:3.10-slim 0.1s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                       0.0s
=> [1/7] FROM docker.io/library/python:3.10-slim@sha256:f5d029fe39146b08200bcc73595795ac19b85997ad0e5001a02c7c32e8769efa 0.1s
=> => resolve docker.io/library/python:3.10-slim@sha256:f5d029fe39146b08200bcc73595795ac19b85997ad0e5001a02c7c32e8769efa 0.1s
=> [internal] load build context                                  0.0s
=> => transferring context: 3.10kB                                    0.0s
=> CACHED [2/7] WORKDIR /app                                       0.0s
=> CACHED [3/7] COPY app.py .                                       0.0s
=> CACHED [4/7] COPY spam_classifier_model.pkl .                  0.0s
=> CACHED [5/7] COPY tfidf_vectorizer.pkl .                       0.0s
=> [6/7] COPY templates ./templates                               0.1s
=> [7/7] RUN pip install --no-cache-dir flask scikit-learn joblib 22.6s
=> exporting to image                                             19.7s
=> => exporting layers                                              16.7s
=> => exporting manifest sha256:bbb4de65d30b56d7c597e1402a4da64b94bdc557406f7f55c758210cfa9c7cfc 0.0s
=> => exporting config sha256:07296caa0f39d84c5039d586ade9c4fbadaffa57d6f6b2e19ded88c426169b5b 0.0s
=> => exporting attestation manifest sha256:d959ac5520a286a33fd019384c6dd97333af967f8e7222ae646a2e10168f9b61 0.1s
=> => exporting manifest list sha256:8b5cb71d6b78c1014ea9e079ea6032980938229fd1133b5f39860bbb1b8a3399 0.0s
=> => naming to docker.io/library/spam-web-app-pes2ug23cs044:latest 0.0s
=> => unpacking to docker.io/library/spam-web-app-pes2ug23cs044:latest 2.7s

(akshay@ AKSHAY) - [~/PES2UG23CS044/ml-docker-PES2UG23CS044]
$
```

```
(akshay@ AKSHAY) - [~/PES2UG23CS044/ml-docker-PES2UG23CS044]
$ docker run -p 5000:5000 spam-web-app-pes2ug23cs044
/usr/local/lib/python3.10/site-packages/sklearn/base.py:442: InconsistentVersionWarning: Trying to unpickle estimator SVC from version 1.2.2 when using version 1.7.2. This might lead to breaking code or invalid results. Use at your own risk. For more info please refer to: https://scikit-learn.org/stable/model_persistence.html#security-maintainability-limitations
warnings.warn(
/usr/local/lib/python3.10/site-packages/sklearn/base.py:442: InconsistentVersionWarning: Trying to unpickle estimator TfidfTransformer from version 1.2.2 when using version 1.7.2. This might lead to breaking code or invalid results. Use at your own risk. For more info please refer to: https://scikit-learn.org/stable/model_persistence.html#security-maintainability-limitations
warnings.warn(
/usr/local/lib/python3.10/site-packages/sklearn/base.py:442: InconsistentVersionWarning: Trying to unpickle estimator TfidfVectorizer from version 1.2.2 when using version 1.7.2. This might lead to breaking code or invalid results. Use at your own risk. For more info please refer to: https://scikit-learn.org/stable/model_persistence.html#security-maintainability-limitations
warnings.warn(
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.17.0.2:5000
Press CTRL+C to quit
172.17.0.1 - - [27/Jan/2026 09:49:55] "GET / HTTP/1.1" 200 -
```

5. SCREENSHOT-5

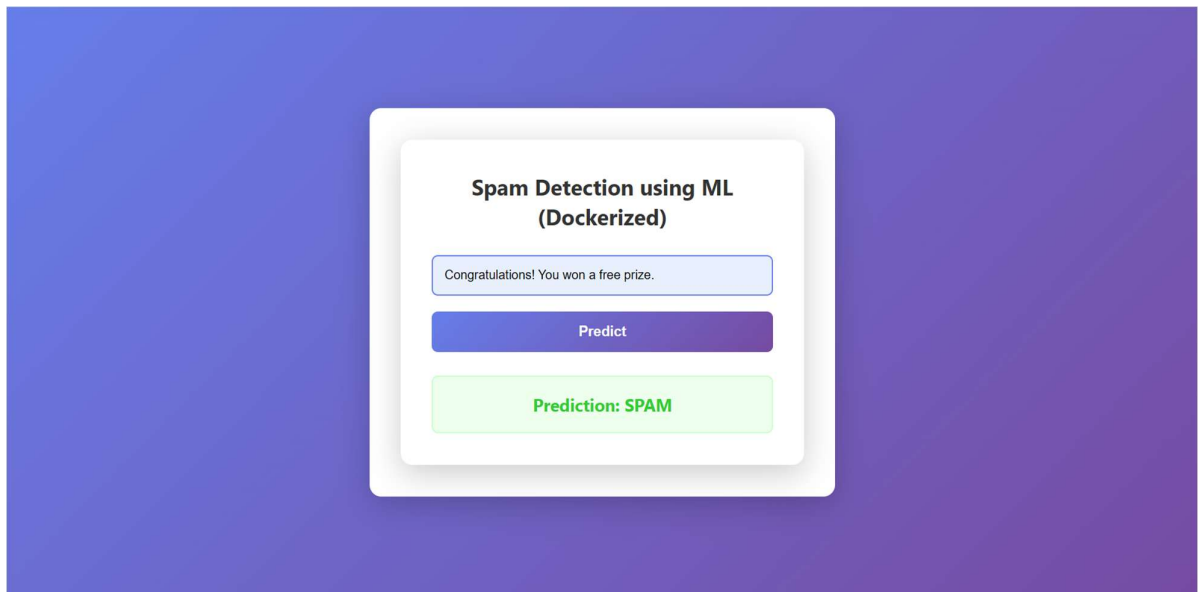
```
(akshay@ AKSHAY) ~/PES2UG23CS044/ml-docker-PES2UG23CS044
$ docker run -d -p 5000:5000 spam-web-app-pes2ug23cs044
3cf756113a3aa156aa345d3d9ca545539129cabf3f7627bf17eef78f19346d33

(akshay@ AKSHAY) ~/PES2UG23CS044/ml-docker-PES2UG23CS044
$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
3cf756113a3a	spam-web-app-pes2ug23cs044	"python app.py"	7 seconds ago	Up 6 seconds	0.0.0.0:5000->5000/tcp, :::5000->5000/tcp	serene_wright

```
(akshay@ AKSHAY) ~/PES2UG23CS044/ml-docker-PES2UG23CS044
$
```

6. OUTPUT SCREENSHOT



7. SCREENSHOT-7

```
(akshay@AKSHAY) - [~/PES2UG23CS044/ml-docker-PES2UG23CS044]
$ docker login
Authenticating with existing credentials...
Login Succeeded
```

8. SCREENSHOT-8

```
(akshay@AKSHAY) - [~/PES2UG23CS044/ml-docker-PES2UG23CS044]
$ docker tag spam-web-app-pes2ug23cs044 akshay/spam-web-app-pes2ug23cs044:v1

(akshay@AKSHAY) - [~/PES2UG23CS044/ml-docker-PES2UG23CS044]
$ docker images
```

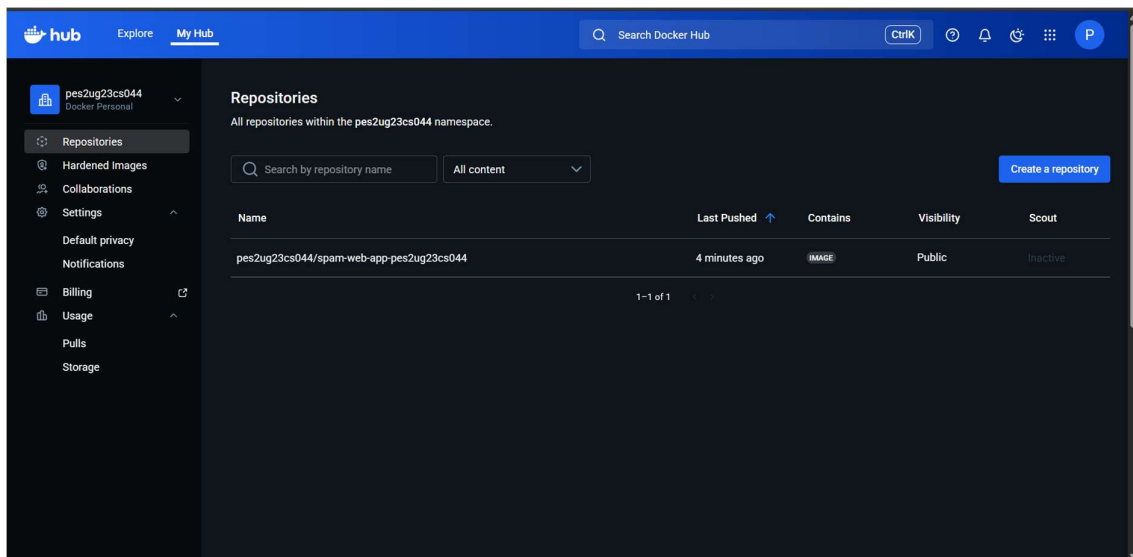
REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
akshay/spam-web-app-pes2ug23cs044	v1	8b5cb71d6b78	14 minutes ago	541MB
spam-web-app-pes2ug23cs044	latest	8b5cb71d6b78	14 minutes ago	541MB
flask-ml-app	latest	359560b72b9b	28 minutes ago	541MB
python	3.10-slim	f5d029fe3914	2 weeks ago	185MB
hello-world	latest	05813aedc15f	5 months ago	25.9kB

```
(akshay@AKSHAY) - [~/PES2UG23CS044/ml-docker-PES2UG23CS044]
$ |
```

9. SCREENSHOT-9

```
(akshay@AKSHAY) - [~/PES2UG23CS044/ml-docker-PES2UG23CS044]
$ docker push pes2ug23cs044/spam-web-app-pes2ug23cs044:v1
The push refers to repository [docker.io/pes2ug23cs044/spam-web-app-pes2ug23cs044]
119d43eec815: Mounted from library/python
47c2bec0e44b: Mounted from library/python
65eac35864ab: Pushed
80ae44bec572: Pushed
90de24bc1bec: Pushed
8e1ed914984e: Pushed
af233a02f79e: Mounted from library/python
49e6a4105476: Pushed
ed8cfd59ed51: Mounted from library/python
285531d0a7aa: Pushed
e42cccf88078: Pushed
v1: digest: sha256:8b5cb71d6b78c1014ea9e079ea6032980938229fd1133b5f39860bbb1b8a3399 size: 856
```

10. SCREENSHOT-10



11. SCREENSHOT-11

```
(akshay@AKSHAY) ~/PES2UG23CS044/ml-docker-PES2UG23CS044
$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS                               NAMES
3cf756113a3a   spam-web-app-pes2ug23cs044         "python app.py"         26 minutes ago Up 26 minutes  0.0.0.0:5000->5000/tcp, :::5000->5000/tcp  serene_wright

(akshay@AKSHAY) ~/PES2UG23CS044/ml-docker-PES2UG23CS044
$ docker stop 3cf756113a3a
3cf756113a3a

(akshay@AKSHAY) ~/PES2UG23CS044/ml-docker-PES2UG23CS044
$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS        NAMES
```

12. SCREENSHOT-12

```
(akshay@AKSHAY) ~/PES2UG23CS044/ml-docker-PES2UG23CS044
$ docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS              PORTS                               NAMES
3cf756113a3a   spam-web-app-pes2ug23cs044         "python app.py"         28 minutes ago Exited (137) About a minute ago    serene_wright
fd66e6a74496   spam-web-app-pes2ug23cs044         "python app.py"         33 minutes ago Exited (0) 28 minutes ago          loving_rhodes
be4b44acf012   affb2574eb17                       "python app.py"         40 minutes ago Exited (0) 35 minutes ago          vibrant_poitras
0acf4ea1d81b   flask-ml-app                       "python app.py"         47 minutes ago Exited (0) 43 minutes ago          dreamy_feynman
014c3a9ef275   hello-world                         "/hello"                 About an hour ago Exited (0) About an hour ago          great_darwin
2494e30e04eb   hello-world                         "/hello"                 About an hour ago Exited (0) About an hour ago          wonderful_sammet

(akshay@AKSHAY) ~/PES2UG23CS044/ml-docker-PES2UG23CS044
$ docker rm 3cf756113a3a
3cf756113a3a

(akshay@AKSHAY) ~/PES2UG23CS044/ml-docker-PES2UG23CS044
$ docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS              PORTS                               NAMES
fd66e6a74496   spam-web-app-pes2ug23cs044         "python app.py"         34 minutes ago Exited (0) 29 minutes ago          loving_rhodes
be4b44acf012   affb2574eb17                       "python app.py"         41 minutes ago Exited (0) 35 minutes ago          vibrant_poitras
0acf4ea1d81b   flask-ml-app                       "python app.py"         48 minutes ago Exited (0) 44 minutes ago          dreamy_feynman
014c3a9ef275   hello-world                         "/hello"                 About an hour ago Exited (0) About an hour ago          great_darwin
2494e30e04eb   hello-world                         "/hello"                 About an hour ago Exited (0) About an hour ago          wonderful_sammet

(akshay@AKSHAY) ~/PES2UG23CS044/ml-docker-PES2UG23CS044
$
```


13. SCREENSHOT-13

```
(akshay@AKSHAY) - [~/PES2UG23CS044/ml-docker-PES2UG23CS044]
$ docker images
REPOSITORY          TAG          IMAGE ID          CREATED           SIZE
spam-web-app-pes2ug23cs044  latest      8b5cb71d6b78     44 minutes ago   541MB
flask-ml-app          latest      359560b72b9b     58 minutes ago   541MB
python               3.10-slim   f5d029fe3914     2 weeks ago      185MB
hello-world          latest      05813aedc15f     5 months ago     25.9kB

(akshay@AKSHAY) - [~/PES2UG23CS044/ml-docker-PES2UG23CS044]
$ docker rmi spam-web-app-pes2ug23cs044:latest
Untagged: spam-web-app-pes2ug23cs044:latest
Deleted: sha256:8b5cb71d6b78c1014ea9e079ea6032980938229fd1133b5f39860bbb1b8a3399

(akshay@AKSHAY) - [~/PES2UG23CS044/ml-docker-PES2UG23CS044]
$ docker images
REPOSITORY          TAG          IMAGE ID          CREATED           SIZE
flask-ml-app        latest      359560b72b9b     58 minutes ago   541MB
python              3.10-slim   f5d029fe3914     2 weeks ago      185MB
hello-world         latest      05813aedc15f     5 months ago     25.9kB

(akshay@AKSHAY) - [~/PES2UG23CS044/ml-docker-PES2UG23CS044]
$ |
```

14. SCREENSHOT-14

```
(akshay@AKSHAY) - [~/PES2UG23CS044/ml-docker-PES2UG23CS044]
$ docker images
REPOSITORY          TAG          IMAGE ID          CREATED           SIZE
spam-web-app-pes2ug23cs044  latest      8b5cb71d6b78     38 minutes ago   541MB
pes2ug23cs044/spam-web-app-pes2ug23cs044  v1         8b5cb71d6b78     38 minutes ago   541MB
flask-ml-app          latest      359560b72b9b     52 minutes ago   541MB
python               3.10-slim   f5d029fe3914     2 weeks ago      185MB
hello-world          latest      05813aedc15f     5 months ago     25.9kB

(akshay@AKSHAY) - [~/PES2UG23CS044/ml-docker-PES2UG23CS044]
$ docker rmi pes2ug23cs044/spam-web-app-pes2ug23cs044:v1
Untagged: pes2ug23cs044/spam-web-app-pes2ug23cs044:v1

(akshay@AKSHAY) - [~/PES2UG23CS044/ml-docker-PES2UG23CS044]
$ docker images
REPOSITORY          TAG          IMAGE ID          CREATED           SIZE
spam-web-app-pes2ug23cs044  latest      8b5cb71d6b78     38 minutes ago   541MB
flask-ml-app          latest      359560b72b9b     52 minutes ago   541MB
python               3.10-slim   f5d029fe3914     2 weeks ago      185MB
hello-world          latest      05813aedc15f     5 months ago     25.9kB

(akshay@AKSHAY) - [~/PES2UG23CS044/ml-docker-PES2UG23CS044]
$ |
```

PES2UG23CS044

//ANSWER

|

15. SCREENSHOT-15

```
(akshay@AKSHAY)-[~/PES2UG23CS044/ml-docker-PES2UG23CS044]
$ docker pull pes2ug23cs044/spam-web-app-pes2ug23cs044:v1
v1: Pulling from pes2ug23cs044/spam-web-app-pes2ug23cs044
8e1ed914984e: Pull complete
49e6a4105476: Pull complete
65eac35864ab: Download complete
Digest: sha256:8b5cb71d6b78c1014ea9e079ea6032980938229fd1133b5f39860bbb1b8a3399
Status: Downloaded newer image for pes2ug23cs044/spam-web-app-pes2ug23cs044:v1
docker.io/pes2ug23cs044/spam-web-app-pes2ug23cs044:v1
```